.sova: Evidence-Bounded Semantic Reproduction of Observable AI Behavior Across Runtimes

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2026-08-03

Abstract

Reproducing observable AI behavior across models and runtimes requires more than byte playback. We introduce an experimental .sova AI-behavior capsule and an evidence-bounded semantic-reproduction decision that couples typed observable outcomes, an execution-context equivalence envelope, evidence-sufficiency checks, multi-valued decisions, and recomputable repeated-trial statistics. Trace playback, controlled re-execution, and semantic reproduction remain distinct operations; the method records observable interactions and never claims access to hidden model reasoning. On a frozen same-team corpus of 144 constructed pairs, the executable mechanism matched all labels while simple structural baselines achieved macro-F1 between 0 and 0.111. A preregistered Wilson-interval rule failed its worst-cell coverage gate (0.914); a disclosed amendment using an exact interval reached 0.956. A 12-case, 24-trace no-Atlas study completed two-backend execution, packaging, inert playback, comparison, and offline verification. On eleven retrospective cross-model pairs derived from 22 AgentRewardBench WebArena trajectories, the mechanism returned two equivalent, five divergent, and four inconclusive decisions. A post-result graph baseline matched 5/7 conclusive labels, and a calibrated semantic-text baseline matched 7/7 but could not represent inconclusive cases. These results establish implementability and falsifiable boundaries, not independent cross-runtime validity. A later frozen public-contract study matched all expected decisions for 12 contracts and 96 traces, accepted zero incomplete traces as reproduced, found zero exact byte matches, and conservatively kept all 12 aggregates inconclusive because each retained two mandatory-evidence gaps.

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Candidate arXiv classification: cs.AI (cross-list candidates: cs.SE, cs.CR)

1. Research question

Can a portable AI-behavior artifact make cross-runtime reproduction decisions that remain:

1. explicit about the observable behavior being tested;
2. sensitive to safety, incompatibility, missing evidence, and invalid provenance;
3. tolerant only to declared non-material variation;
4. recomputable from retained per-trial evidence; and
5. more informative than exact trace, final-output, event-kind, or terminal- status equality?

The falsifiable hypothesis is that the complete contract will reduce false semantic equivalence and false semantic divergence against these baselines on held-out, independently labeled cases. The current synthetic corpus tests implementation consistency, not the final hypothesis.

2. Artifact model

`.sova` is a lightweight, versioned, inert-by-default AI-behavior capsule. A capsule can package or reference scenarios, triggers, environments, dependencies, models, agents, tools, traces, fixtures, assertions, evaluations, findings, hypotheses, provenance, integrity, citation, disclosure, and authorship records.

`.sova-trace` is the canonical low-level observable event and evidence stream. The capsule organizes material required to inspect, replay, reproduce, compare, study, verify, or cite a behavior. It is not a claim that a model's private thoughts were captured.

Three operations are disjoint:

- **trace playback** renders an inert recorded timeline;
- **controlled re-execution** performs a fresh authorized run and creates a new trace;
- **semantic reproduction** evaluates whether a declared material outcome recurs under an explicit context and evidence contract.

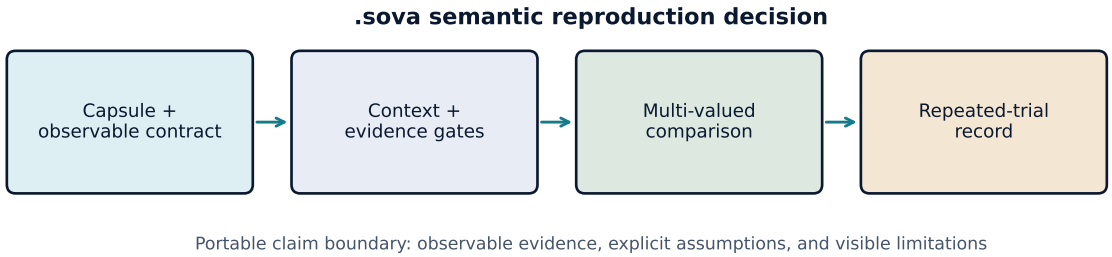


Figure 1: Conceptual architecture for Evidence-Bounded Semantic Reproduction. Boxes show the portable decision boundary; executor or producer mechanics remain outside the claim.

3. Proposed mechanism

3.1 Typed observable outcome contract

A contract contains required and optional predicates over observable event families. The current deterministic prototype supports existence, absence, equality, containment, minimum count, and ordering. Each predicate identifies the event family, value path, expected relation, and evidence events used.

A missing required event family produces `unknown`, not `fail` or `pass`. Optional future judges must record identity, version, inputs, configuration, calibration, uncertainty, and disagreement. The integrated contract now supports typed human/model/rule adjudication records with pseudonymous adjudicator identity, label, blinding and implementation-independence assertions, integer confidence basis points, and private-rationale digests. It reports unanimous, plurality, tie, or absent agreement and never silently overrides the evidence-derived outcome. This implements the record structure; it does not create independent reviewers or validate their assertions.

3.2 Execution-context equivalence envelope

Each context dimension declares one comparison mode:

- `exact`: values must match;
- `one-of`: the candidate must belong to an allowed set;
- `tolerance`: a numeric deviation is bounded;
- `record`: divergence is recorded but does not block comparison;
- `ignore`: the dimension is explicitly non-material.

Missing required context produces an unknown comparison. This prevents a receiver from silently treating an absent model, tool, protocol, or dependency version as equivalent.

3.3 Evidence-bound decision

The decision state is one of:

- `equivalent`;
- `divergent`;
- `inconclusive`;
- `incompatible`;

- `unsafe`;
- `evidence-invalid`.

Safety and evidence validity are checked before behavior predicates. Incompatible context is not counted as ordinary behavioral failure. An otherwise matching behavior is inconclusive when required context equivalence is unknown.

3.4 Reproduction-rate record

Every trial retains:

- trial identity;
- outcome state;
- environment stratum;
- evaluator identity;
- evidence digest.

The summary reports counts for every state, the success rate among conclusive trials, Wilson and exact binomial intervals, and a conservative partial- identification range in which inconclusive trials may be either success or failure. Incompatible, unsafe, and invalid-evidence trials remain visible but are not silently folded into a Bernoulli denominator.

Correlated runs must be grouped and modeled rather than counted as independent replicates. A production study should use a hierarchical or generalized linear mixed model when model, item, environment, and judge variation warrant it.

4. Relationship to prior work

The design does not claim novelty for portable archives, traces, workflow replay, deterministic replay, provenance, semantic evaluation, behavioral contracts, repeated trials, or statistical intervals.

ReproEvalCard (Pattnayak and Bhatia 2026) specifies reproducibility-critical material for LLM evaluation pipelines. `agrepl` (Mudasiru 2026) records and deterministically replays external interactions. WebGraphEval (Qian et al. 2025) compares agent trajectories through canonical graphs. ReliableEval addresses stochastic and prompt-sensitive evaluation. Semantic-Eval studies semantic generation metrics. ContractEval (Lim et al. 2026) and Phoenix (Wang and Huang 2026) demonstrate explicit behavioral-contract evaluation. PoC-Adapt (Duy et al. 2026) uses structured semantic oracles for vulnerability reproduction. W3C PROV (World Wide Web Consortium 2013) defines normalization, validity, and equivalence for provenance instances. NIST AI 800-2 and 800-3 (National Institute of Standards and Technology 2025; Keller et al. 2025) require explicit uncertainty assumptions and motivate hierarchical statistical models.

The candidate contribution is therefore the narrower coupling of observable outcome semantics, context equivalence, evidence sufficiency, multi-valued reproduction states, and recomputable uncertainty inside a portable cross- runtime behavior artifact.

5. Experimental protocol

The protocol was frozen privately before corrected-mechanism results.

5.1 Synthetic behavior corpus

- six profiles: security, evaluation, agent trajectory, behavioral interpretability, incident forensics, and research publication;
- 24 cases per profile, 144 total;
- balanced ground-truth classes across equivalent, divergent, inconclusive, incompatible, unsafe, and evidence-invalid;
- typed label, causal-order, and forbidden-egress predicates;
- context drift in model family, runtime, protocol, temperature, and host.

5.2 Baselines

1. exact canonical trace equality;
2. exact final observable payload equality;

3. terminal completion-status equality;
4. normalized request/outcome interaction equality;
5. untyped event-kind equality.

These are intentionally simple lower baselines, not substitutes for direct integration with WebGraphEval, semantic metrics, or full replay systems.

5.3 No-Atlas execution study

Twelve owned deterministic conditional fixtures alternated trigger and control inputs. Each capsule ran through `ScriptedExecutor` and `RestrictedLocalExecutor`, yielding 24 traces. Each case required:

- successful bounded execution;
- passing observable oracle;
- declared-outcome equivalence across executor mechanics;
- native and dependency-free offline trace verification;
- inert playback;
- an evidence capsule containing scenario plus both traces.

No network, external credential, live target, Atlas MCP, or model provider was used.

5.4 Post-result integrated and Inspect baseline study

After the preregistered study, protocol amendment 0.5 connected the private mechanism to actual public `.sova` packages, verified `.sova-trace` evidence, both no-Atlas executors, deterministic migration, an inert renderer, and a private CLI. Seven controlled cases covered cross-executor equivalence, repeat equivalence, required-context incompatibility, an unsafe comparison, a partial prefix, provider success contradicted by the SOVA oracle, and a complete run lacking mandatory evidence.

The same traces were converted to AgentEvals 0.0.9's documented OpenAI-style message/tool-call representation and evaluated by its deterministic strict, unordered, and superset trajectory matchers with exact tool arguments. The package is MIT licensed and the evaluation used no model, key, or network call. This is a mature-library structural-trajectory baseline, but the seven labels and conversion remain same-team.

A separate adapter was pinned to Inspect AI (UK AI Security Institute 2026) repository commit `1ea01a9e1b3c94db3cdf93ce5583ffebf86e159f` and EvalLog version 2. It converted a bounded synthetic JSON-dump subset into SOVA events with an explicit fidelity report. Four cases varied surface output, source score, and score availability. The adapter does not parse Inspect's binary `.eval` format and the initial direct-baseline experiment did not use a third-party real log.

5.5 Pinned external corpora and implementation baselines

The Inspect adapter was run against every JSON EvalLog in four frozen official test-log directories at repository commit `1ea01a9e1b3c94db3cdf93ce5583ffebf86e159f`. The thirteen-file corpus contains transcript events, legacy flat events, no-event-stream records, and logs with no samples. Adapter 0.2 accepted five files and typed-refused eight because it assumed `sample.transcript`. That negative result was retained. Adapter 0.3 added explicit layout detection and was rerun against the unchanged corpus. Raw official fixture bytes are not redistributed.

The AgentRewardBench (Lu et al. 2025) authors' official demo Space at commit `271f96552e2e6dd13fe8e3572f0e5264039` contained twenty-two cleaned WebArena trajectories: two real-model agent families over eleven shared task identifiers. Every trajectory was converted into a verified trace/capsule using only task/model provenance, action verb/digest, and grouped expert success, side-effect, optimality, and looping labels. Goal text, raw reasoning, action arguments, screenshots, accessibility trees, URLs, chat messages, and annotator identity were excluded and probed in decoded retained structures. The two models were paired by task under the predeclared multi-valued rule.

Harness Evals 0.15.0 supplied three separately maintained deterministic baselines over the unchanged seven-case constructed corpus: `ExactMatch`, `ToolArgumentMatch`, and `ToolCorrectness`. The study used no model, account, credential, provider, or network call.

The Space metadata did not assert a source-data licence, so source bytes are not redistributed. The trajectory study is retrospective and the SOVA interpretation remains same-team. It does not establish controlled re-execution, environment equivalence, blind SOVA-specific annotation, or independent review.

5.6 Supported-runtime repetition

The research tests, integrated baseline, and no-Atlas vertical study were executed in isolated environments on CPython 3.11.15, 3.12.13, 3.13.12, and 3.14.3. Volatile timestamps and performance measurements were excluded before comparing the stable decision slices. All runtimes executed on one Windows host.

5.7 Frozen Topic 12 public-contract study

After the Topic 12 implementation was complete, a separate protocol froze 12 observable contracts. Each contract was evaluated against eight deterministic candidate traces: four equivalent-runtime variations, two material-outcome variations, and two traces missing mandatory evidence. The expected per-trace decision vector was fixed as four `reproduced`, two `not-reproduced`, and two `inconclusive`. The study also required zero exact byte matches, zero incomplete traces accepted as reproduced, and an aggregate result that did not hide the missing-evidence trials.

The public implementation uses Wilson intervals and sensitivity strata. It is not the same as the private Clopper-Pearson and partial-identification research prototype. This study therefore validates an implemented conservative subset without converting the private candidate into a public novelty claim.

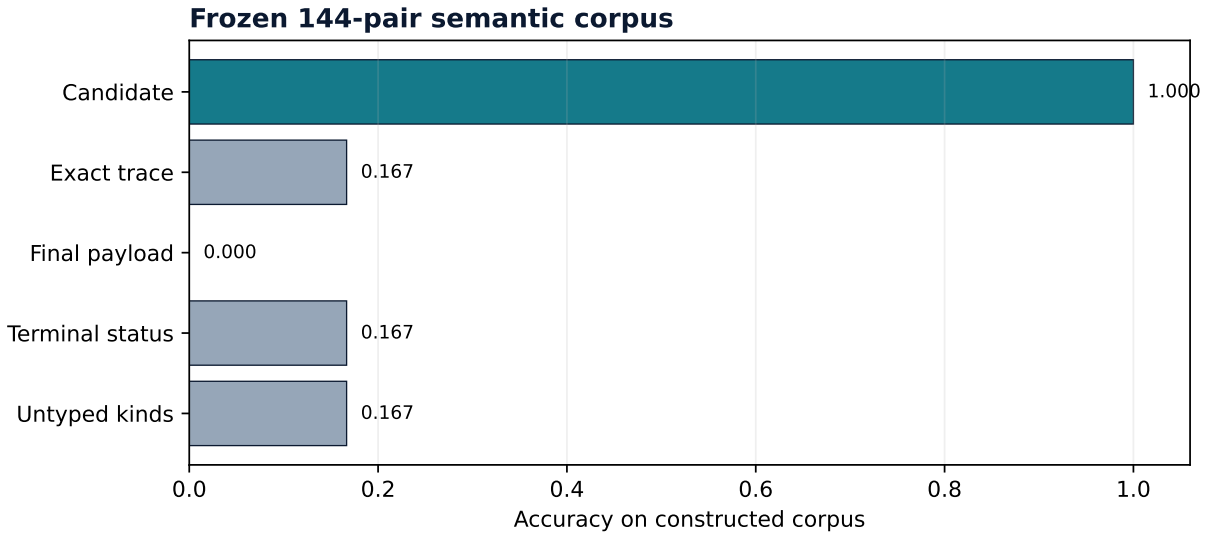


Figure 2: Constructed-corpus baseline comparison. Values are retained experiment results, not field-performance estimates.

6. Results

6.1 Classification

Method	Macro-F1
Contracted mechanism	1.000
Exact trace	0.111
Exact final payload	0.000
Terminal status	0.048
Untyped event kinds	0.048
Remove context envelope	0.778
Remove evidence-validity gate	0.778
Collapse inconclusive into failure	0.778
Collapse inconclusive into success	0.778

The contracted mechanism made zero false-equivalence decisions and zero false divergence decisions on the frozen constructed corpus. Its full confusion matrix is retained in the result artifact. Because this prototype emits categorical decisions rather than probabilistic confidence scores, confidence calibration

is not applicable; a future learned or model-based judge must add an independently labeled calibration study.

The perfect score is expected to be optimistic because the same team designed the corpus transformations and the evaluator. It is evidence of correct implementation against declared construction rules, not external validity. The post-result ablations each misclassified the one balanced class whose distinction they removed, reducing macro-F1 to 0.778. They were added under protocol amendment 0.3 and are not represented as preregistered analyses.

6.2 Uncertainty gate and negative result

The initial preregistration required nominal 95% Wilson coverage. The seeded simulation's aggregate coverage passed, but the stricter worst-cell interpretation failed. Analytic Wilson coverage reached only 0.914 in the worst predeclared `(p,n)` cell.

The original result was retained. Protocol amendment 0.2 added a two-sided Clopper-Pearson interval computed by binomial-tail inversion. Minimum analytic coverage over the same grid was 0.956. The exact interval is more conservative; the statistical method is established prior art and is not a contribution.

6.3 No-Atlas vertical slice

All 12 cases and 24 traces passed the predeclared implementation checks. Scripted and local executors used different mechanics and deliberately differed in non-material stdout formatting, while the declared containment predicate remained equal. Both traces and the derived capsule passed the independent dependency-free verifier.

6.4 Integrated direct-baseline results

Method	Correct / total	Accuracy
Integrated evidence-bounded mechanism	7 / 7	1.000
Public declared-observable exact comparator	5 / 7	0.714
Normalized request/outcome interaction equality	3 / 7	0.429
Terminal completion equality	2 / 7	0.286
AgentEvals 0.0.9 strict trajectory match	2 / 7	0.286
AgentEvals 0.0.9 unordered trajectory match	2 / 7	0.286
AgentEvals 0.0.9 superset trajectory match	2 / 7	0.286
AgentEvals 0.0.9 graph-strict trajectory match	2 / 7	0.286
Exact trace transport bytes	1 / 7	0.143

The public comparator correctly handled ordinary observable equality, divergence, and missing selected evidence, but it could not represent an unsafe comparison or a required-context incompatibility. The integrated mechanism also forced `inconclusive` when a complete trace lacked mandatory claim evidence; before this test, behavioral predicates and fidelity results were computed but not conservatively composed. That defect was corrected and retained in the falsification record.

The normalized interaction comparator improved over completion and byte equality by comparing request action/inputs and terminal status/output. It still could not represent unsafe-to-attempt or incompatible contexts and forced incomplete evidence into binary divergence. This is a controlled request-key baseline inspired by transport replay, not a direct execution of `agrepl` and not proof of transport isolation.

All four AgentEvals modes accepted the equivalent pair and repeat pair. They also accepted structurally equal interactions under incompatible context, an unsafe comparison, and a provider success contradicted by the SOVA oracle; they called both missing-terminal/evidence cases divergent rather than inconclusive. This does not show AgentEvals is defective for its intended structural-matching use. It shows why structural trajectory matching alone is not the same decision problem as evidence-bounded semantic reproduction.

On four pinned synthetic Inspect JSON-import cases, the candidate matched all four constructed labels. Exact final-output equality matched one and terminal completion equality matched two. All imported traces were explicitly marked lossy. These tiny same-team cases demonstrate adapter implementability

and a counterexample to output equality; they do not establish performance against Inspect, WebGraphEval, semantic metrics, or real evaluation logs. The later thirteen-log Inspect corpus validates bounded format handling, while the twenty-two-trajectory AgentRewardBench cohort supplies eleven retrospective cross-model pairs with external labels. Neither is controlled re-execution or purpose-built blind SOVA annotation.

Two independent offline implementations, one Python and one dependency-free Node.js, agreed on package/content digests and required-key DSSE verification for the same trace. This is cross-language engineering evidence, not an independent scientific replication.

6.5 External deterministic baselines

Method	Correct / total	False equivalence
Evidence-bounded candidate	7 / 7	0
Harness `ExactMatch`	3 / 7	2
Harness `ToolArgumentMatch`	2 / 7	3
Harness `ToolCorrectness`	2 / 7	5

The Harness methods do not natively represent the incompatible, inconclusive, or unsafe truth classes in this corpus. These results support the narrower claim that binary structural matchers are insufficient for the frozen multi-valued evidence problem. They do not show that Harness is defective for its intended evaluation tasks or that SOVA is generally superior.

6.6 External real-model retrospective cohort

All twenty-two AgentRewardBench WebArena trajectories produced verified traces and capsules, with zero known forbidden-content findings in the decoded retained structures. Seven trajectories contained expert-label disagreement. The eleven cross-model task pairs produced:

Outcome	Count
Equivalent	2
Divergent	5
Inconclusive	4

The equivalent rate among seven conclusive comparisons was 2/7. Its two-sided Wilson interval was 0.0822-0.6411; treating all four inconclusive cases as unknown yielded a partial-identification interval of 0.1818-0.5455 over all eleven pairs.

Retrospective baseline	Exact agreement with SOVA decision
Expert success-only	7 / 11
Exact action sequence	6 / 11
Action multiset	6 / 11
Terminal action	2 / 11

The frequent inconclusive outcome is not a defect to be optimized away: it records external expert disagreement rather than manufacturing equivalence. The cohort is too small and selectively exposed in a demo Space to support a population or state-of-the-art claim.

6.7 Cross-version decision stability

The integrated stable decision slice had digest `sha256:891c07912f28e7743ea145593e73aa3d78c5e42ec10fc665cd110134` under every tested CPython minor version. The no-Atlas vertical decision slice had digest `sha256:f27cd1c4fdcff9903bf89e41de9436dc9e930b34ffc0347b16c8fd2f494d6f6f` under every version. This is one-host cross-version evidence, not independent cross-machine replication.

6.8 Graph and calibrated semantic-text baselines

The post-result baseline study pinned WebGraphEval's published construction as the conceptual graph reference but deliberately did not reproduce its LLM canonicalizer or necessity annotations. Instead, it built a deterministic two-pass edit-similarity action graph with a 0.9 merge threshold. On the eleven retrospective AgentRewardBench pairs it predicted every pair as divergent. It therefore matched 5/7 conclusive labels and missed both equivalent pairs; it also had no representation for the four inconclusive labels.

The semantic baseline used `model2vec==0.8.2` and the MIT-licensed `minishlab/potion-base-8M` model pinned at `bf8b056651a2c21b8d2565580b8569da283cab23`. Platt calibration used the STS Benchmark train split and no test refit. On the held-out STS test split it obtained accuracy 0.8216, Brier score 0.1267, ten-bin expected calibration error 0.0382, Pearson 0.7726, and Spearman 0.7541. With the explicitly post-result 0.5 trajectory threshold it matched 7/7 conclusive AgentRewardBench labels, but it again forced all four inconclusive cases to divergent.

This is useful negative evidence: graph or embedding similarity can recognize some material equivalence, but neither baseline carries authorization, environment compatibility, evidence sufficiency, or an inconclusive state. The semantic result is exploratory because calibration data are ordinary sentence pairs rather than agent trajectories and threshold selection occurred after the original P4 studies. No raw sentences, goals, URLs, reasoning, or actions are retained in the result artifact.

6.9 Topic 12 public-contract result

All 12 contracts matched their complete expected decision vectors across 96 candidate traces. Decision accuracy was 1.000. The implementation accepted zero incomplete traces as reproduced, and exact transport bytes matched zero candidate traces. All 12 aggregate reports were `inconclusive`: each contained four reproduced of six eligible trials plus two mandatory-evidence gaps. The equivalent-runtime stratum reported 4/4 reproduced, the material-drift stratum 0/2, and the missing-evidence stratum zero eligible plus two inconclusive.

This is an intentionally adverse result for a headline success rate. It shows that the implementation preserves uncertainty rather than silently dropping incomplete trials. The fixtures and expected vectors remain same-team and rule-derived, so the result is implementation evidence only.

7. Threats to validity

Construct validity

The typed predicates may omit behavioral meaning important to a domain. Passing them does not prove that the scenario author chose the correct contract. The current corpus does not test open-ended semantic judges.

Internal validity

The research team constructed both labels and implementation. The perfect synthetic score may reflect specification alignment. The next study requires independent labelers and hidden cases.

External validity

Four CPython minor versions on one Windows reference host, deterministic fixtures, thirteen Inspect logs, and twenty-two retrospective WebArena trajectories do not represent controlled provider drift, fresh stochastic trials, browser-state equivalence, distributed tools, or real incidents. Hosted CI provides platform engineering evidence, not behavioral generalization.

Statistical conclusion validity

Synthetic repetitions do not justify population-level claims. Trial dependence and environment effects require an explicitly fitted model in a real study.

Security validity

`RestrictedLocalExecutor` is not a security sandbox. The study contains no offensive payload and establishes no target security property.

8. Future independent validation

The current evidence is same-team and bounded. The next confirmatory study must use independently authored behavior pairs, blinded labels, at least two independently operated runtimes, controlled real-model re-execution, and predeclared comparisons with the strongest applicable graph, semantic, and record/replay baselines. Unsafe cases must remain refusals, and incompatible or missing-evidence cases must not be collapsed into binary failure. The external-review packet freezes the cases, digests, release order, and scoring rules but cannot manufacture reviewer independence.

9. Conclusion

The .sova capsule makes an AI behavior a portable, inspectable research object while keeping execution mechanics and evidentiary claims explicit. The retained experiments show that the proposed decision is executable and expose where byte equality, terminal status, graph similarity, and semantic similarity answer narrower questions. The mechanism should be treated as a testable architecture, not as established superiority, until independent cross-runtime validation supports or falsifies it.

Declarations

Ethics and dual-use statement

The reported machine experiments use synthetic or public research fixtures and safe local actions. They do not target a live third party, use client data, store real credentials, or claim access to private model thoughts or hidden chain-of-thought. The work can support security testing and therefore has dual-use potential. Unsafe reproduction remains subject to explicit authorization, containment, and human review; operational exploit details and novelty-sensitive implementations are not part of this private review package.

Data and artifact availability

The generic SOVA implementation, public specifications, safe examples, and tests are available in the SOVA-OSS repository. Private experiment results, preregistrations, source digests, build provenance, and review packets are retained by TRUSCOR Private Limited. Third-party source artifacts whose redistribution rights are unclear are not redistributed. A release artifact must be assembled only after rights, privacy, security, and IP review.

Author contributions and AI-assisted work

Gautam R. Patil conceived the SOVA project, specified the research direction and safety constraints, and directed this work. Software implementation, experiment execution, analysis support, literature discovery, and drafting used OpenAI Codex under his direction. Gautam R. Patil is the sole proposed human author and must verify the final text, citations, figures, data, and venue disclosure before submission. No AI system is listed as an author or inventor.

Funding and competing interests

This work was conducted for TRUSCOR Private Limited. No external funding was used for the reported machine experiments. The author's affiliation with TRUSCOR Private Limited is a relevant non-financial and potential commercial interest. The exact legal entity name and any additional interests must be confirmed before submission.

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